TopMark Tutors

SACE STAGE 2 · REVISION CHECKLIST

Biology

Every topic you need to know, in one checklist.

4 topics | 45 checkpoints | TopMark Tutors

How to use this checklist

Work through every line and tick honestly. Do the whole thing in one sitting before you revise anything — that first pass becomes your study plan for the term.

1 — Revised
You have gone back over the content and understand it.

2 — Practised
You have done real exam questions on it, not just read notes.

3 — Confident
You could do it in the exam, under time, with no notes.

Most students stop at column 1. Re-reading notes feels like studying, but marks are won in columns 2 and 3. Anything ticked in column 1 and blank in column 3 is exactly where your marks are leaking.

Externally assessed by an ELECTRONIC (online) examination — practise on screen, not just on paper.

Topic 1 — DNA and Proteins

DNA structure and replication

CONTENT	REVISED	PRACTISED	CONFIDENT
Structure of DNA and RNA, and differences between them	☐	☐	☐
DNA replication, including the role of key enzymes	☐	☐	☐
Chromosome structure and the organisation of genetic material	☐	☐	☐

Gene expression

Transcription and translation, step by step	☐	☐	☐
Using a codon table accurately (a common source of lost marks)	☐	☐	☐
Types of mutation and their effects on the resulting protein	☐	☐	☐
Regulation of gene expression and the influence of environment	☐	☐	☐

Proteins and enzymes

Amino acid structure and levels of protein structure	☐	☐	☐
Enzyme action, active site and specificity	☐	☐	☐
Effect of temperature, pH and concentration on enzyme activity	☐	☐	☐
Interpreting enzyme rate graphs	☐	☐	☐

Biotechnology

DNA profiling and gel electrophoresis	☐	☐	☐
PCR and its applications	☐	☐	☐
Genetic modification techniques, plus benefits, risks and ethical implications	☐	☐	☐

WEAK SPOTS TO COME BACK TO

Topic 2 — Cells as the Basis of Life

TOPIC 2 OF 4

Cell structure and transport

CONTENT	REVISED	PRACTISED	CONFIDENT
Prokaryotic vs eukaryotic cells, and organelle structure and function	☐	☐	☐
Membrane structure (fluid mosaic model)	☐	☐	☐
Diffusion, osmosis, facilitated diffusion and active transport	☐	☐	☐
Surface area to volume ratio and its consequences for cell size	☐	☐	☐

Cellular energy

Photosynthesis: light-dependent and light-independent stages, and where each occurs	☐	☐	☐
Cellular respiration: glycolysis, Krebs cycle and electron transport chain	☐	☐	☐
Correctly locating each stage within the cell (frequently examined)	☐	☐	☐
Aerobic vs anaerobic respiration and comparative ATP yield	☐	☐	☐

Cell division

The cell cycle and its checkpoints	☐	☐	☐
Mitosis vs meiosis, and clearly identifying where they differ	☐	☐	☐
Crossing over, independent assortment and sources of genetic variation	☐	☐	☐
Cancer as uncontrolled cell division	☐	☐	☐

WEAK SPOTS TO COME BACK TO

Topic 3 — Homeostasis

Principles of homeostasis

CONTENT	REVISED	PRACTISED	CONFIDENT
Negative feedback loops: stimulus, receptor, control centre, effector, response	☐	☐	☐
Writing a complete feedback loop without skipping a step	☐	☐	☐
Tolerance limits and the consequences of exceeding them	☐	☐	☐

Control systems

Structure and function of the nervous system, and nerve impulse transmission	☐	☐	☐
The endocrine system and mechanisms of hormone action	☐	☐	☐
Comparing nervous and endocrine control (speed, duration, specificity)	☐	☐	☐

Regulation in the body

Thermoregulation in endotherms and ectotherms	☐	☐	☐
Osmoregulation and kidney function, including ADH	☐	☐	☐
Glucoregulation: insulin, glucagon and diabetes	☐	☐	☐
Regulation of blood pressure and the role of adrenaline	☐	☐	☐

WEAK SPOTS TO COME BACK TO

Topic 4 — Evolution

Evidence and mechanisms

CONTENT	REVISED	PRACTISED	CONFIDENT
Evidence for evolution: fossils, comparative anatomy, molecular evidence	☐	☐	☐
Natural selection and the conditions required for it to occur	☐	☐	☐
Distinguishing natural selection from selective breeding and genetic drift	☐	☐	☐
Allele frequency change within populations	☐	☐	☐

Speciation and biodiversity

Reproductive isolating mechanisms (pre-zygotic and post-zygotic)	☐	☐	☐
Allopatric and sympatric speciation	☐	☐	☐
Interpreting phylogenetic trees and cladograms	☐	☐	☐
Human impact on biodiversity and conservation strategies	☐	☐	☐
Antibiotic and pesticide resistance as evolution in action	☐	☐	☐

WEAK SPOTS TO COME BACK TO

WHAT TO DO WITH THE BLANKS

The gaps you just found are the whole point.

Every line you could not tick in the ‘Confident’ column is a mark you are currently at risk of dropping. Our SACE Exam Crash Course is built to close exactly those gaps — we mapped four years of real SACE papers to find where marks are actually lost, then weighted the teaching toward those points instead of re-covering content you already know.

The SACE Exam Crash Course — Biology

- 28 September – 10 October · Methods, Chemistry, Physics, Biology
- 12 hours of teaching per subject · online or in person, same live sessions
- TopMark Biology textbook included free
- Finishes with a full mock exam, marked and returned with your personal mark-loss breakdown
- 32 five-star Google reviews · 100% of our 2025 cohort scored above a 90 ATAR

From $349 for one subject, or $20/hr across all four. A $99 deposit secures your spot. Early bird pricing ends 6 September.

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